LESSON PLAN

HIGH SPEED FLIGHT (12)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction, propeller function, tractor and pusher types, material need and construction of propeller
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction, propeller function, tractor and pusher types, material need and construction of propeller
5. Specific Objectives	12.1.1.1 Trainee will be able to describe the various types of propellers, principles of operation and pitch change mechanisms.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of propeller diagrams; Safety: No handling of rotating parts; Supervision: Instructor monitors diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a propeller?
(b) Name two common types of propeller configurations.
(c) Why is material selection critical for propeller construction?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Propeller function and thrust generation • Tractor and pusher configuration types • Blade material requirements and properties • Basic construction and assembly methods	L & Ds	PPT, IB	15 min
(b)	• Principles of propeller operation • Fixed pitch versus variable pitch • Pitch change mechanisms and control • Aerodynamic efficiency factors	L & Ds	PPT, PDF	20 min
(c)	• Construction details of hub and blades • Balancing and dynamic stability • Maintenance considerations for materials • Integration with engine power output	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between tractor and pusher propeller types.
(b) Describe how a variable pitch mechanism alters blade angle.
(c) Identify key material properties required for propeller blades.

12. Home Assignment and Follow Up	Prepare a brief note on the advantages of feathering and braking mechanisms to be discussed in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Definition of terms, power absorption, feathering & braking & its advantages.	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min